Johan Lakshmanan

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Education

University of Massachusetts, Amherst Amherst, MA
Bachelor of Science in Computer Science — 3.7 (Deans List) *Aug. 2023 – May 2026*

Experience

Roblox Jan 2026 – Present
Incoming Software Engineering Intern *San Mateo, CA*

- Selected for the Cell Lifecycle team to build scalable backend services and infrastructure tools that manage the end-to-end lifecycle of Roblox's global compute clusters.
- Collaborating on mission-critical systems to ensure seamless performance and low-latency connectivity for a global user base of **70M+**.

Fidelity Investments Jun. 2025 – Aug. 2025
Quantitative Research Intern *Boston, MA*

- Piloted a model-driven trade router, scaling trade volume by **1.4x** and saving **$200K+ over 300+** hedge funds.
- Developed and integrated backend machine learning models in Python and Kdb+/Q to optimize routing for **25M+** trades, improving selection accuracy by 20% and reducing decision latency below **10 ms**.
- Constructed automated training, deployment, and data pipelines on AWS EKS using Jenkins CI/CD with containerized microservices and MLflow, supporting reproducible model versioning and rollback in production.

Waters Corporation May 2024 – Aug. 2024
Software Engineering Intern *Milford, MA*

- Engineered a full-stack automated renewal quote system using **Python**, RESTful APIs, and a responsive UI, reducing manual workload by **100+ hours monthly**, streamlining sales operations.
- Designed multiple **REST APIs** to retrieve customer SharePoint resources by environment ID, decreasing mitigation times from **hours to minutes** for runtime incidents, reducing loads for on-call engineering teams.
- Developed high-concurrency C# modules to interface with **SAP S/4HANA**, optimizing SQL query execution and reducing transaction latency by **25%**.

Build UMass Feb. 2024 – May. 2024
Software Engineer *Amherst, MA*

- Led full-stack development of a consulting platform using MERN stack, deployed on **Dockerized AWS EC2** with Nginx load balancing and **GitHub Actions** CI/CD.
- Rebuilt RESTful APIs with pagination, Redis caching, and profiling, reducing latency by **40%** and supporting **5x** higher concurrent request volumes.
- Partnered with designers to improve UI responsiveness and reduce load times by **20% for 200+** concurrent users.

MIT Lincoln Laboratory May 2021 – Aug. 2022
Embedded Systems Intern *Cambridge, MA*

- Built a secure bootloader for Stellaris microcontrollers (C/Assembly) and designed penetration testing frameworks (Python/SQL), patching **16+ vulnerabilities** and earning **1st place** in the MITLL Cybersecurity Challenge.

Projects

AI Bitcoin Sentiment Platform | *Python, FastAPI, SQLAlchemy, NLP, Docker* Aug. 2025 – Present

- Built a **modular ETL pipeline** aggregating cryptocurrency news, Reddit, and price feeds.
- Designed and deployed a FastAPI microservice for scheduled ingestion, FinBERT-based sentiment inference, and caching, optimizing inference latency by **120ms** and ensuring **95%+** uptime for scheduled data ingestion.

Skills

Languages: Python, Java, C/C++, TypeScript, JavaScript, SQL, HTML/CSS
Infrastructure & DevOps: AWS (EC2, EKS), Docker, Kubernetes, Terraform, Jenkins, Github Actions, Nginx
Frameworks: React, Node.js, Express, Django, Redis, FastAPI, Pandas, Material-UI, RESTful APIs
Technologies: Git, Linux, VMware, JIRA, VS Code

Awards

Collegiate Penetration Testing Competition — **3rd Place (Global Finals)** Jan. 2024